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Predicting Gene Ontology functions based on support vector machines and statistical significance estimation

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Abstract

Gene Ontology (GO) is a common language for the functional annotation of gene products. We have developed a computational tool, GOKey, to predict the GO function of proteins based on their sequence features and the support vector machine (SVM) method. Several measures, including improved handling of the problem caused by unbalanced positive and negative training data and postprocessing strategies to evaluate the posterior probability and statistical significance of SVM outputs, have been adopted to improve the prediction performance of GOKey. The GOKey has been trained to predict the 36 GO categories of the 'molecular function' of GO slims, and could be easily extended to other GO categories. The results of 5-fold cross validation with 10,603 GO-mapped proteins demonstrate that the performance of GOKey is better than that of standard SVMs. Comparisons with other computational tools for GO function prediction also show that the performance of GOKey is satisfactory. Further, GOKey has been applied to predict the GO functions for 5381 novel human proteins in the Ensembl database. The results show that 93% of the novel proteins can be assigned one or more GO terms, and some evidences supporting the predictions have been found. GOKey can be accessed at http://infosci.hust.edu.cn.

Keywords: Protein function; Gene Ontology; Support vector machines; Statistical significance

1. Introduction

The characterization of biochemical functions of gene products is an important task in the postgenomic era. With the progress in genome sequencing of more and more model organisms, however, there exists an increasing gap between the amount of available sequence data and the experimental characterization of their functions. Accurate computational function prediction, that is helpful for speeding up the functional annotation of gene products, has become an increasingly important problem in the field of bioinformatics [1–3].

Gene Ontology (GO), a set of structured vocabularies organized in a rooted directed acyclic graph (DAG), can provide consistent description of gene products in a species-independent manner [4,5], and has become a common language for the functional annotation of gene products. In recent years, many computational methods for the functional annotation of gene products have been developed based on GO, which can be classified as context based methods and sequence based methods according to the utilized information [1].

High-throughput experimental context data such as the protein–protein interaction and gene expression data provide a valuable resource, for the prediction of protein functions. The representatives of context-based methods include the Markov random field model to exploit protein–protein interaction data [6,7] and the clustering analysis to make use of gene expression data [8]. Although these methods are promising and have achieved certain success, currently, their usefulness is still limited by the quality and quantity of the data [9,10].

It is widely accepted that proteins' functions are determined, to a great extent, by their three-dimensional structures and the structures are determined by the amino acid sequences. Therefore, how to predict protein functions from sequence information is essential and attractive. The representatives of sequence-based computational tools for GO function prediction include GoFigure [11],

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GOblet [12], OntoBlast [13], ProtFun [14] and GO™ Predictor [15]. The GoFigure, GOblet and OntoBlast are based on homologous information, which predict the GO terms of uncharacterized protein sequences with the GO annotations of similar sequences found in GO-mapped protein databases [11–13]. The ProtFun is an artificial neural network-based program, which can predict specific GO terms by using sequence derived protein features such as predicted sorting signals, post-translational modifications and some physical/chemical properties [14]. The GO™ Predictor, which is based on the assumption that protein functions are determined by domains, makes predictions using detected domains and a set of previously generated rules for domain-function associations [15].

Recently, a relatively new classification method, support vector machines (SVMs) [16,17], has been introduced to solve various biological problems such as microarray data analysis [18], protein fold recognition [19], protein–protein interaction prediction [20], protein secondary structure prediction [21] and protein function prediction [22,23]. Specifically, Cai et al. [22] studied how to use the SVM method and protein sequence information to predict protein function, and developed a web-based tool SVMProt for classifying proteins into such functional classes as enzymes, receptors, transporters, channels, binding proteins and so on [22,23]. However, these protein function terms, written in rich and non-formalized language, are context dependent, and the vagueness in using such terms may yield confusing annotations [3,24,25]. In this work, we have developed a computational tool, GOKey, to predict the GO functions of proteins based on their sequence features and the SVM method. Especially, on the basis of SVM predictions, several postprocessing measures have been adopted to improve the prediction performance of GOKey.

2. Materials and methods

2.1. The configuration of GOKey

The configuration of our function predictor GOKey is shown in Fig. 1. The GOKey consists of a coding unit and a calculation unit. The calculation unit, which contains a group of SVMs (each SVM corresponding to a specific GO category) and other postprocessing models for estimating the probabilities (the 'probability' modules) and statistical significance (the '*P*-value' modules) of the prediction results of SVMs, is the kernel of the system.

In principle, the GOKey can be trained to predict any GO term for which there are enough training data. Here, we adopt the 'molecular function' of the GO slims (updated on 7 December 2004), which have been applied in the annotation of several genomes [5], as an example to clarify the application of GOKey. Of the 40 nodes in 'molecular function' of GO slims, three nodes, GO:0030188 (chaperone regulation activity), GO:0045735 (nutrient reservoir activity) and GO:0030533 (triplet codon-amino

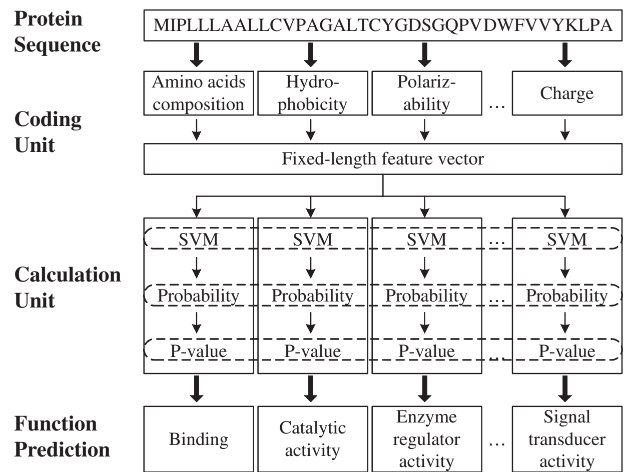

Fig. 1. The configuration of GOKey system.

acid adaptor activity) are discarded because they currently cover very few proteins (less than ten protein sequences) to build effective prediction models. In addition, GO:0005554 (molecular function unknown) is also excluded. The process for developing the GOKey that can predict the remaining 36 GO terms includes the following steps:

(1) Selecting positive and negative samples for each of the 36 GO categories. The positive samples are a set of protein sequences annotated with the GO term, whereas the negative samples are a set of sequences that do not have the corresponding GO function. The positive and negative samples are used as the benchmark data for constructing models that can effectively distinguish proteins with a specific GO function from those without.
(2) Encoding positive and negative samples using sequence features. The purpose of this step is to construct feature vectors of positive and negative samples using sequence features for building classification models.
(3) Building classification models using feature vectors. The goal of protein function prediction is to decide which GO category an uncharacterized protein belongs to. Obviously, this is a classification problem. There exist many classification methods that first constructing decision functions (classifiers) with a training dataset and then make decisions according to the values of these functions (decision values). In this study, the SVM and several postprocessing models are adopted.
(4) Validating the classification models. The performance of the classification models is evaluated with the positive and negative samples by a 5-fold cross-validation process.

2.2. Selection of positive and negative samples

To select reliable positive and negative samples for each of the 36 GO nodes, here, we use the concept of functional path of GO nodes. The functional path of a GO node is the path on the DAG of GO that can reach that node from the root. Therefore, if a protein is annotated with a GO node, it also belongs to all the nodes on its functional paths. For each GO node, all proteins that have been annotated with this node or its descendant nodes are considered as the positive samples of this GO node. Since the current functional annotation might be incomplete, it is hard to justify whether or not the proteins annotated with the nodes on the functional paths of a GO node belong to this GO node, and therefore these proteins can only be considered as unlabelled samples of the GO node. Then, all the proteins excluding the positive samples and unlabelled samples of a GO node are designated as the negative samples of the GO node.

The positive and negative samples are extracted from SWISS-PROT database (release 44.5) [26]. Only sequences that are annotated with non-IEA evidence and are not marked with 'fragment' are considered. The number of positive and negative samples for each of the 36 GO nodes is listed in Table 1.

2.3. Encoding positive and negative samples

In GOKey, each protein sequence is first transformed into a fixed-length feature vector by the coding unit and then processed by the calculation unit. The feature vectors of protein sequences are assembled from encoded representations of tabulated residue properties including the amino acids composition, hydrophobicity, normalized van der Waals volume, polarity, polarizability and charge. The schemes of how to convert tabulated residue properties into encoded representations are the same as that used in the references [19,22,23].

Table 1
List of GO classes, statistics of datasets and cross-validation results of the SVM method and GOKey

GO ID	Positive	Negative	SVM method			GOKey[a]			GOKey[b]		
			Sn (%)	*Sp* (%)	*MCC*	*Sn* (%)	*Sp* (%)	*MCC*	*Sn* (%)	*Sp* (%)	*MCC*
GO:0016209	48	10,555	37.5	100	0.60	50.0	100	0.67	70.8	99.8	0.68
GO:0005488	4950	5653	69.5	75.8	0.45	71.2	74.6	0.46	75.1	70.3	0.45
GO:0005509	154	10,421	25.3	100	0.50	37.7	99.9	0.56	46.8	99.5	0.52
GO:0030246	66	10,521	22.7	100	0.48	39.4	100	0.58	43.9	99.9	0.56
GO:0008289	85	10,502	0	100	NA	7.1	100	0.20	16.5	99.8	0.27
GO:0003676	1682	8905	30.7	98.8	0.46	38.5	97.9	0.49	59.5	92.0	0.51
GO:0003677	1085	9485	24.1	99.1	0.40	33.2	98.7	0.46	53.3	95.6	0.51
GO:0003682	46	10,098	8.7	100	0.29	21.7	100	0.44	28.3	99.8	0.34
GO:0003700	501	9631	12.2	99.8	0.28	25.5	99.3	0.39	47.7	97.9	0.48
GO:0003723	539	10,031	29.1	100	0.52	40.3	99.7	0.59	52.5	98.8	0.59
GO:0008135	78	10,484	2.6	100	0.16	24.4	100	0.46	32.1	99.9	0.44
GO:0000166	264	10,323	10.6	100	0.32	25.8	99.9	0.46	43.2	99.0	0.47
GO:0019825	22	10,565	0	100	NA	31.8	100	0.43	63.6	99.5	0.37
GO:0005515	2005	8582	17.9	98.7	0.31	24.4	97.5	0.34	41.7	91.9	0.37
GO:0008092	186	9222	13.4	100	0.36	25.3	99.9	0.46	41.4	99.3	0.47
GO:0003779	93	9282	18.3	100	0.43	24.7	100	0.47	40.9	99.7	0.48
GO:0005102	657	9813	27.2	99.9	0.50	43.8	99.5	0.59	59.5	97.6	0.59
GO:0003824	3760	6843	68.8	88.0	0.58	67.1	88.7	0.58	72.6	86.3	0.59
GO:0016787	1351	9218	11.1	100	0.31	26.4	99.0	0.42	42.3	95.4	0.43
GO:0004518	107	10,441	0	100	NA	4.7	100	0.18	19.6	99.8	0.30
GO:0008233	380	10,172	10.8	100	0.32	30.8	99.7	0.49	43.4	98.9	0.49
GO:0004721	132	10,416	5.3	100	0.23	22.0	99.9	0.41	34.1	99.5	0.39
GO:0016740	1327	9240	9.3	99.9	0.27	28.3	98.4	0.41	46.4	94.2	0.43
GO:0016301	688	9877	9.0	100	0.29	27.6	99.4	0.44	49.4	96.7	0.47
GO:0004672	533	10,015	11.6	100	0.33	37.1	99.5	0.53	56.5	97.3	0.52
GO:0030234	499	10,104	2.8	100	0.16	12.0	99.9	0.31	25.7	98.8	0.34
GO:0003774	78	10,525	19.2	100	0.41	32.1	99.9	0.48	46.2	99.6	0.47
GO:0004871	1973	8630	40.2	98.9	0.55	50.3	97.7	0.59	64.4	93.3	0.59
GO:0004872	1092	9388	46.7	99.4	0.63	55.6	98.7	0.65	64.2	97.6	0.67
GO:0005198	564	10,039	29.1	99.9	0.52	38.3	99.7	0.57	48.2	98.9	0.57
GO:0030528	1071	9532	27.5	98.8	0.41	35.1	98.3	0.46	55.6	95.9	0.53
GO:0045182	89	10,514	2.2	100	0.15	21.3	100	0.44	28.1	99.8	0.41
GO:0005215	1195	9408	34.1	99.5	0.52	42.8	99.0	0.57	51.1	97.6	0.57
GO:0005489	188	10,330	6.4	100	0.25	17.0	99.9	0.36	22.9	99.6	0.34
GO:0005216	203	10,310	10.3	100	0.32	41.9	99.8	0.58	48.8	99.7	0.60
GO:0005326	16	10,502	0	100	NA	12.5	100	0.29	50.0	99.8	0.36
Overall	27,707	34,9577	37.5	99.2	0.52	45.7	98.9	0.56	57.9	97.5	0.58

[a]Posterior probability 0.5 is used as cutoff.
[b]*P*-value 0.005 is used as cutoff.

2.4. The SVM

The SVM is a standard supervised learning algorithm, which builds a hyper plane to separate the positive and negative samples in a multiple-dimensional space (original space). However, most practical problems involve non-separable data that cannot be successfully separated with a hyper plane in the original space. One solution to this problem is to map the data in original space into a higher-dimensional space and define a separating hyper plane there. This higher-dimensional space is called the feature space. The SVM method uses a kernel function of the vectors in original space to define the dot product of the vectors in feature space, that can avoid a lot of trouble caused by the high dimensionality of the feature space. Recent studies have suggested the usefulness of SVMs in solving biological problems such as microarray data analysis, protein fold recognition, prediction of protein–protein interaction, prediction of protein structure and function etc. For details of SVM, please refer to Refs. [16,17].

In this study, The LIBSVM (http://www.csie.ntu.edu.tw/~cjlin/libsvm/) program is used as the implementation of SVM, in which the radical basis function kernel is adopted and the two parameters C and γ are empirically set to 1 and 10, respectively.

2.5. The estimation of posterior probability

The outputs generated by the decision function of a standard SVM are uncalibrated values. However, constructing a classifier that can output posterior probabilities is very useful for predicting protein function. For example, posterior probabilities are required to combine heterogeneous data to predict protein functions [27]. Here, we use Platt's probabilistic outputs for SVMs, that are popular for applications that require posterior class probabilities [28]. The posterior probability $p(y=1|\mathbf{x})$ is approximated by a sigmoid function

$$p(y=1|\mathbf{x}) = \frac{1}{1+\exp(Af(\mathbf{x})+B)}, \tag{1}$$

where $\mathbf{x}$ represents the feature vector of a protein sequence and y denotes its class label ($+1$ for positive samples and -1 for negative samples). $f(\mathbf{x})$ is the decision function of SVM. A and B are two parameters determined by maximum likelihood estimation from a training set (f_i, y_i). First, a new training set (f_i, t_i) is generated, where t_i denotes the target probability defined as $t_i=(y_i+1)/2$. Then the parameters A and B are estimated by minimizing the negative log likelihood of the training data, which is a cross-entropy error function

$$\min -\sum_i t_i \log(p_i) + (1-t_i)\log(1-p_i), \tag{2}$$

where

$$p_i = \frac{1}{1+\exp(Af_i+B)}. \tag{3}$$

In machine learning, when there is great disparity between the size of positive and negative training samples, the training difficulty called the imbalance problem must be taken into consideration [29]. This problem is serious for the prediction of GO functions, because there are definitely much more negative functional members than positive ones for each GO node. Here, the out-of-sample model [28] is used to tackle this problem. Assume that there are N^+ positive samples and N^- negative samples, then the estimation of target probability of positive samples is $t^+=(N^++1)/(N^++2)$, while the estimation of target probability of negative samples is $t^-=1/(N^-+2)$.

2.6. The estimation of statistical significance

The posterior probability indicating prediction confidence is an effective measure for the determination of whether or not a protein sequence belongs to a GO category. When the posterior probability is greater than 0.5, the target sequence is more likely to belong to the category. However, there still remains the question whether this prediction result is statistically significant or not. Here, we address this issue by determining the statistical significance value (P-value).

The P-value is calculated from the bit score S. The bit score, a log-odds score in base two, is defined as

$$S = \log_2 \frac{p(y=1|x)}{1-p(y=1|x)}, \tag{4}$$

where $p(y=1|\mathbf{x})$ is defined by expression (1). For a target sequence whose bit score is x, its P-value is the probability $P(S \geqslant x)$ that the bit score S of any random sequence is higher than or equal to x. Because there is no statistical theory that allows for direct computation of the probability of obtaining a certain score by chance, one has to rely on empirical methods for the estimation of statistical significance [30]. The first step of these methods is to construct a random sequence database. For this purpose, we shuffled all the 10,603 protein sequences in our dataset. The resulted random sequence database has the additional advantage that certain statistical properties of protein sequences can be preserved. The score distribution of these random sequences is then analyzed by using the chi-square goodness-of-fit test, which shows that the extreme value distribution (EVD) fits the score distribution very well. The distribution function of the EVD is

$$P(S \geqslant x) = 1 - \exp\left[-e^{-\lambda(x-\mu)}\right], \tag{5}$$

where the parameters λ and μ are determined by maximum likelihood estimation.

3. Results

3.1. Cross-validation results

The GOKey program has been trained with the collected positive and negative samples, and can be used to predict the 36 GO categories of the 'molecular function' of GO slims mentioned in Section 2. P-value 0.005 is the default cutoff value used in GOKey. Here, 5-fold cross-validation method is adopted to examine the prediction performance of the SVM method and GOKey for each of the 36 GO categories. For comparison, the performance of GOKey when using posterior probability 0.5 as the cutoff value is also tested. Three measures, the prediction sensitivity Sn, specificity Sp and the Matthews correlation coefficient MCC which can describe the overall prediction performance [23], are applied to evaluate the performance. Let TP, TN, FP and FN denote the number of true positives, true negatives, false positives and false negatives, respectively, then Sn, Sp and MCC can be defined as $Sn = TP/(TP+FN)$, $Sp = TN/(TN+FP)$, $MCC = (TP \times TN - FP \times FN)/((TP+FP)(TP+FN)(TN+FP)(TN+FN))^{1/2}$. The results of 5-fold cross-validation for each of the 36 GO categories are given in Table 1.

From Table 1, it can be seen that the Matthews correlation coefficients (representing overall performance) of GOKey are significantly better than those of SVM method for most GO categories. Comparison between the results of GOKey with different cutoff values shows that P-value criterion can increase the prediction sensitivity significantly, and get better overall performance although the specificity drops a little.

3.2. Comparison with the GOA electronic annotation

GOA provides a GO mapping for human proteome dataset [31]. The comparison results between the predictions of GOKey and the GOA electronic annotation for molecular function are summarized in Fig. 2. Of the 48,953 protein sequences for *Homo sapiens* (IPI_HUMAN v3.03), 39% are assigned GO functions by both GOKey predictions and GOA electronic annotations. Within this set of proteins, 78% agree on at least one GO term according to the DAG structure of GO. For 55% of the 48,953 proteins, GOKey can predict their GO categories but there is no function annotation from GOA. In contrast, there are only 3% of proteins that GOA can provide function annotations but GOKey cannot. The results demonstrate that GOKey can achieve a certain consistency with GOA electronic annotation and cover more proteins than GOA electronic annotation.

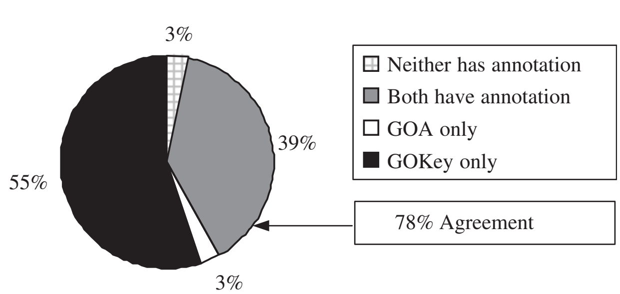

Fig. 2. Comparison with the GOA electronic annotation for human proteome. The chart shows the percentages of proteins for *Homo sapiens* that have been assigned GO functions by GOA electronic annotation and/or by GOKey. In cases where both methods have yielded assignments, the figure indicates the percentage of proteins whose annotations have some agreement.

3.3. GO function prediction for novel proteins

In Ensembl, protein models that cannot be mapped to species-specific entries in SWISS-PROT, RefSeq or TrEMBL are referred to as novel proteins [32]. There are 5381 such novel proteins for *Homo sapiens* in the Ensembl database (release 29.35b), and we have tried to predict their GO functions with GOKey. The results show that 5004 out of the 5381 novel proteins (93%) can be assigned one or more GO terms by GOKey (P-value < 0.005). Table 2 lists the top 50 proteins that are assigned GO functions with high confidence. The posterior probabilities for these predictions are approximately equal to 1, and the P-values are lower than 10^{-9}.

To demonstrate the usefulness of these predictions, we have manually inspected these entries of Ensembl and found some evidences that support these predictions. For example, the evidence that the protein ENSP00000266376 is a subunit of voltage-dependent calcium channel confirms the prediction of GOKey that this protein possesses the ion channel activity (GO:0005216). The evidence that the protein ENSP00000316037 has an eIF-5A domain (eukaryotic initiation factor 5A hypusine) supports the prediction that this protein has the function of GO:0008135 (translation factor activity, nucleic acid binding). The evidence that the protein ENSP00000349945 is a member of the olfactory receptor family supports the prediction that this protein displays receptor activity (GO:0004872). Actually, except for ENSP00000309219 and ENSP00000346789, all predictions listed in Table 2 have such supporting information.

4. Discussion

4.1. Comparison with other methods of GO function prediction

So far, little has been done to quantify the accuracy of GO function prediction by standard benchmarks, that is, different datasets were used by different authors to evaluate the performance of their prediction tools. For example, Jensen et al. [14] extracted 12,500 protein sequences from InterPro database as their cross-validation dataset, and Schug et al. [15] used 4357 manually curated human proteins in SWISS-PROT as their test set. Most of

Table 2
Predicted functions for novel human proteins

Protein	Predicted function
ENSP00000266376, ENSP00000349520, ENSP00000351101	GO:0005216 ion channel activity
ENSP00000316037, ENSP00000309323	GO:0008135 translation factor activity, nucleic acid binding
ENSP00000349945, ENSP00000328742, ENSP00000347003	GO:0004872 receptor activity
ENSP00000345931	GO:0005509 calcium ion binding
ENSP00000282292, ENSP00000329020, ENSP00000330785, ENSP00000351398	GO:0003700 transcription factor activity
ENSP00000311082, ENSP00000332008, ENSP00000339058, ENSP00000342294, ENSP00000346975, ENSP00000351738, ENSP00000352086, ENSP00000352132, ENSP00000352313, ENSP00000354469	GO:0003723 RNA binding
ENSP00000348986	GO:0005102 receptor binding
ENSP00000318153, ENSP00000318568, ENSP00000333272, ENSP00000341744, ENSP00000343583, ENSP00000349627, ENSP00000350602, ENSP00000354280, ENSP00000355222	GO:0008233 peptidase activity
ENSP00000341779	GO:0004721 phosphoprotein phosphatase activity
ENSP00000323106, ENSP00000343658	GO:0004672 protein kinase activity
ENSP00000311687, ENSP00000343374	GO:0005215 transporter activity
ENSP00000285052, ENSP00000311082, ENSP00000314498, ENSP00000330779, ENSP00000332008, ENSP00000334733, ENSP00000345230, ENSP00000346623, ENSP00000346975, ENSP00000349424, ENSP00000349790, ENSP00000351022, ENSP00000351449, ENSP00000351738, ENSP00000352086, ENSP00000352132, ENSP00000352313, ENSP00000354469	GO:0005198 structural molecule activity
ENSP00000309219	GO:0016787 hydrolase activity
ENSP00000346789	GO:0030234 enzyme regulator activity

the sequences in InterPro database are also the entries of SWISS-PORT or TrEMBL, so the dataset used in this study would largely overlap that used by Jensen et al. and would cover Schug's dataset because our dataset consists of all manually annotated proteins in SWISS-PROT. As a result, to some extent, it makes sense to compare GOKey's performance obtained with our dataset with the reported results of other tools.

The ProtFun was developed by Jensen et al. using sequence derived features such as the predicted post-translational modifications, protein sorting signals and so on. So its performance would be limited by the prediction accuracy of these features. In fact, many functional categories cannot be predicted by ProtFun, only 14 GO categories are currently covered by it [14]. In contrast, as can be seen from Table 1, when GOKey is trained to predict the 36 GO categories of GO slims, it is able to predict all of these GO functions with reasonable accuracy. Of the 10,603 tested proteins, 10,116 proteins can be assigned one or more GO functions by GOKey (i.e. the prediction coverage is 95%), and 81% of them have at least one predicted GO function showing agreement with the curated GO annotations (i.e. the prediction agreement is 81%). Furthermore, since our method is quite general, GOKey could be easily extended to cover other GO categories by suitable training processes. To compare the performance of GOKey with that of GO^{TM} Predictor developed by Schug et al. [15], we give in Table 3 their prediction coverage and agreement. It can be seen that the prediction coverage of GOKey is significantly larger than that of GO^{TM} Predictor while their prediction agreement values are almost the same.

4.2. Strengths of the method used in GOKey

The method used in GOKey differs from standard SVMs in two major aspects: the improved handling of the problem caused by unbalanced positive and negative training data (i.e. the imbalance problem) and the post-processing measures to evaluate the statistical significance of SVM outputs.

The advantage of our method in handling the imbalance problem can be clarified by comparing the influence of unbalanced training data on the prediction performance of GOKey with that of the SVM method. Let the imbalance extent denote the ratios of the number of positives to negatives, the relationship between the imbalance extent and the Matthews correlation coefficient MCC (representing overall performance) for each of the 36 GO categories is shown in Fig. 3. The least-square fitting curves of the relationship between the imbalance extent and MCC demonstrate that the influence of imbalance problem on prediction performance is serious for SVM, while the influence is trivial for GOKey. For most GO categories, especially those with serious imbalance problem, the prediction performance of GOKey is significantly better than that of SVM method. These results demonstrate the effectiveness of our method in handling the imbalance problem.

The post-processing measures enable GOKey to evaluate the posterior probability and statistical significance value (P-value) of SVM outputs and make decision based on them. The posterior probability reflects whether a protein is more likely to be a member of specific functional family (for posterior probability higher than 0.5) or not (for

Table 3
Prediction coverage and agreement of GOKey and GO^{TM} predictor [15]

Method	Prediction coverage (%)	Prediction agreement (%)
GOKey	95	81
GO^{TM} predictor	81	82

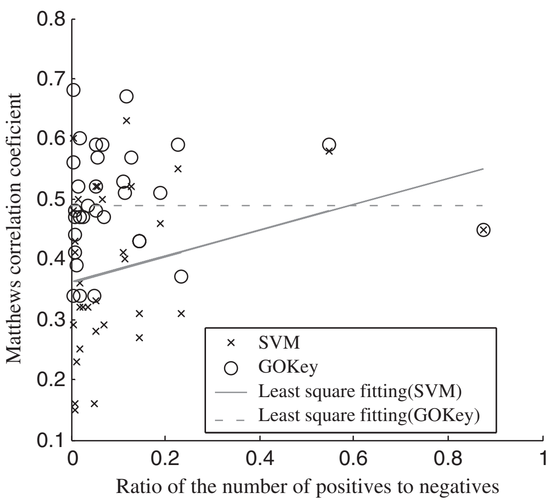

Fig. 3. Relationship between the imbalance extent and the Matthews correlation coefficient for the SVM method and GOKey. Each cross or circle represents a GO category of the 'molecular function' of GO slims.

Table 4
GOKey predictions for protein Q9Y4I1

Protein	GO ID	GO term	Probability	*P*-value
Q9Y4I1	GO:0005488	Binding	0.54	0.005
	GO:0003774	Motor activity	0.14	7.6e−6

posterior probability lower than 0.5), while the *P*-value measures how statistically significant the prediction result is. Usually, A GOKey model for a specific GO category will give both high posterior probability and significant *P*-value for members of this functional family. For distantly related members, however, there exist the case that the probability is low but the *P*-value is significant, indicating that the probability, though low, is still higher than expected by chance, and that the *P*-value as used in GOKey is a more effective measure than the probability. For the example given in Table 4, the protein sequence Q9Y4I1 is predicted to be of two GO categories: GO:0005488 and GO:0003774. Although the posterior probability for GO:0003774 is much lower than 0.5, its *P*-value is significant, suggesting that GO:0003774 is a good prediction. Actually, this prediction is consistent with the annotation in SWISS-PROT. In contrary, the posterior probability for GO:0005488 is higher than 0.5, but its *P*-value is not that significant. In fact, according to current annotations of SWISS-PROT, this GO function has not been assigned to this protein.

5. Conclusions

We have developed a computational tool, GOKey, to predict the GO functions of proteins based on their sequence features and the SVM method. The sequence features used in GOKey include the amino acids composition, hydrophobicity, normalized van der Waals volume, polarity, polarizability and charge, which are the same as that used by others. However, several measures, including the improved handling of the problem caused by unbalanced positive and negative training data and the post-processing strategies to evaluate the posterior probability and statistical significance of SVM outputs, have been adopted to improve the prediction performance. The results of 5-fold cross validation with 10603 GO-mapped proteins demonstrate that the prediction performance of GOKey is better than that of standard SVMs. Comparisons with the prediction results of other computational tools for GO function prediction also show that the performance of GOKey is satisfactory. These results indicate that the methods to estimate the posterior probability and statistical significance of prediction results as well as to handle the problem of unbalanced positive and negative training data are effective to improve the performance of SVM based prediction programs. In addition, although only 36 GO categories are currently covered by GOKey, it could be easily extended to predict other GO categories by suitable training processes. Future works will focus on combining heterogeneous data and refining the selection of positive and negative samples to further improve the prediction performance and extend the coverage of GO categories of the GOKey program.

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